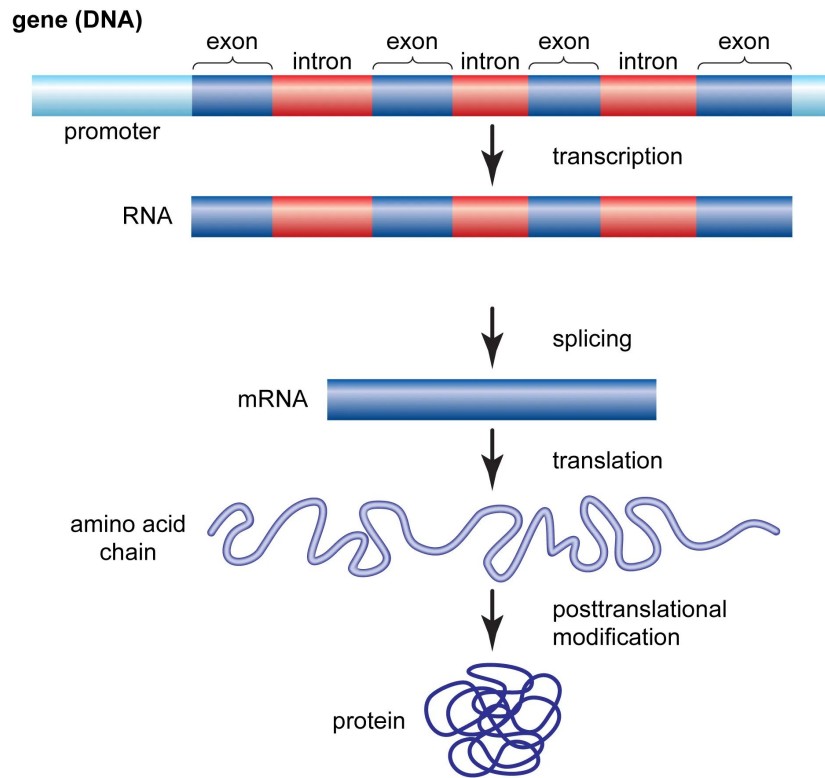


Hidden Markov Models

Adi, Howard, Isabel, James

Motivation

- In biology we have several macromolecules of interest composed of an observable sequence with an underlying, unknown order
- A Hidden Markov Model (HMM) is a flexible tool to extract these hidden states → determine some structure in what may look like randomness

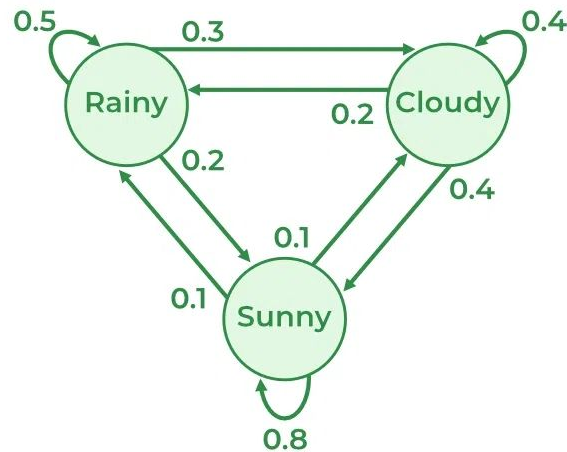


HMM Use Cases in Biology

- Gene prediction (CpG Islands, exon/intron structure)
- Copy number variation detection
- Chromatin domain identification from ChIP-seq
- Protein structure prediction
- Topologically Associating Domains (TAD) identification

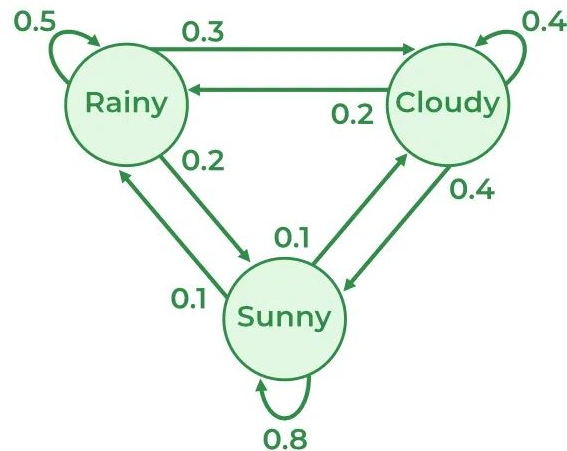
What is a Markov Model? (Stat 110 Review)

- Markov Models encode states and transition probabilities
- Visually represented as a graph with each state as a node and each transition path as an edge
- Markov Property: The future behavior of the stochastic process only depends on the current state, otherwise referred to as “memorylessness”



What is a Markov Model?

- Mathematically represented using matrices with transition probabilities:
 - Matrix A where A_{ij} represents the probability of moving from state i to state j
- We also need some initial state:
 - Probability π_i of starting in state i



↓

	R	C	S
R	0.5	0.3	0.2
C	0.2	0.4	0.1
S	0.1	0.1	0.8

Hidden Markov Model

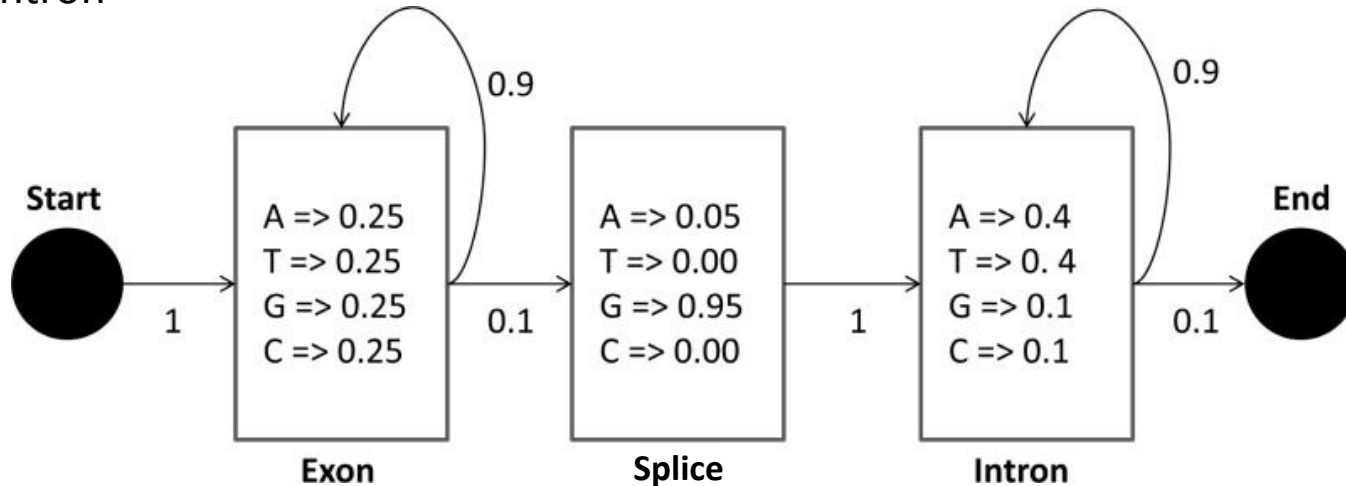
- When the system is in a state, what does it do?
 - Under the HMM structure, it emits certain observable characteristics with certain probabilities
- Now we have “hidden” *transitions* and observed *emissions*
- Now we have a transition matrix A and an emission matrix B
 - A_{ij} is the probability of moving from hidden state i to hidden state j
 - B_{jk} is the probability of observing k when in hidden state j
 - π_i is the probability of starting in state i
- Consider the same weather example, but our only observable is whether the temperature is high (H) or low (L)

$$A = \begin{bmatrix} & R & C & S \\ R & 0.5 & 0.3 & 0.2 \\ C & 0.2 & 0.4 & 0.4 \\ S & 0.1 & 0.1 & 0.8 \end{bmatrix}$$

$$B = \begin{bmatrix} & H & L \\ R & 0.2 & 0.8 \\ C & 0.4 & 0.6 \\ S & 0.7 & 0.3 \end{bmatrix}$$

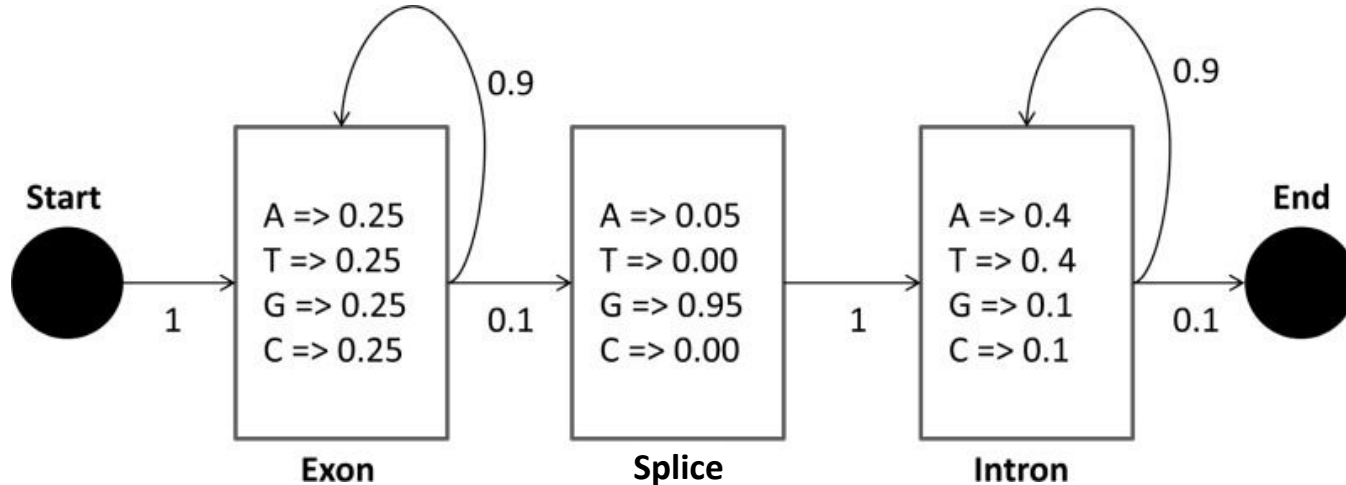
HMM Biology Example: Exon/Intron Prediction

- Introns and Exons tend to have slightly different nucleotide composition due to GC content, codon structure, sequence signals at splice sites
- Say that we want a HMM to predict a single nucleotide G-rich splice site between exon and intron



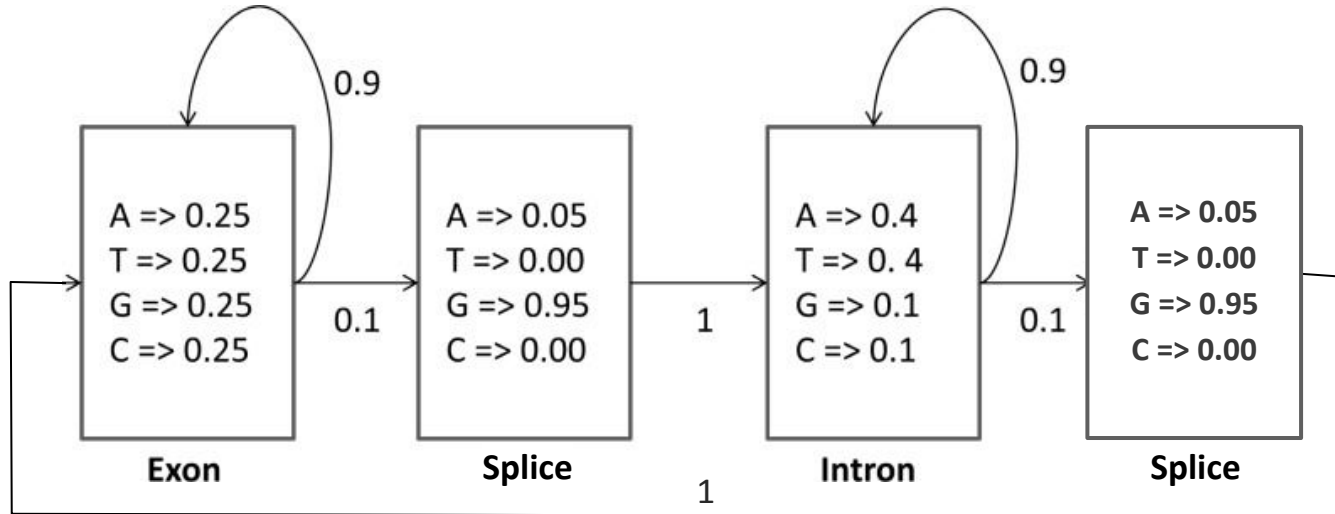
HMM Biology Example: Exon/Intron Prediction

$$A = \begin{bmatrix} & E & 5 & I & \text{End} \\ E & 0.9 & 0.1 & 0 & 0 \\ 5 & 0 & 0 & 1 & 0 \\ I & 0 & 0 & 0.9 & 0.1 \end{bmatrix} \quad B = \begin{bmatrix} & A & T & G & C \\ E & 0.25 & 0.25 & 0.25 & 0.25 \\ 5 & 0.05 & 0.00 & 0.95 & 0.00 \\ I & 0.4 & 0.4 & 0.1 & 0.1 \end{bmatrix} \quad \pi = \begin{bmatrix} E & 5 & I & \text{End} \\ 1 & 0 & 0 & 0 \end{bmatrix}$$



HMM Biology Example: Exon/Intron Prediction

$$A = \begin{bmatrix} & E & 5 & I & 3 \\ E & 0.9 & 0.1 & 0 & 0 \\ 5 & 0 & 0 & 1 & 0 \\ I & 0 & 0 & 0.9 & 0.1 \\ 3 & 1 & 0 & 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} & A & T & G & C \\ E & 0.25 & 0.25 & 0.25 & 0.25 \\ 5 & 0.05 & 0.00 & 0.95 & 0.00 \\ I & 0.4 & 0.4 & 0.1 & 0.1 \\ 3 & 0.05 & 0.00 & 0.95 & 0.00 \end{bmatrix} \quad \pi = \begin{bmatrix} E & 5 & I & 3 \\ 0.49 & 0.01 & 0.49 & 0.01 \end{bmatrix}$$



Path Tracing

- Path tracing is using the HMM (A , B , π) to answer questions about hidden states given observations
- Three algorithms that each answer different questions:
- Viterbi Algorithm: what's the single most likely sequence of hidden states?
- Forward-Backward Algorithm: how likely is each individual position to be in each state?
- Baum-Welch Algorithm: what if I don't know A , B , or π ?

Viterbi Algorithm

- Given a sequence of observations, what's the most likely labeling?
- Naive approach: enumerate every possible labeling. For N positions and k states that's k^N paths
- Viterbi's trick: at each position, only remember the best path ending in each state. Walk left to right, one position at a time
- At each step: for each state, look back one position, pick whichever predecessor gives the highest score
- Markov property says future only depends on current state, so a worse path to the same state can never come back to win
- At the end: trace pointers back to recover full labelling

Biological Applications (Real World)

Sequence & Family Analysis:

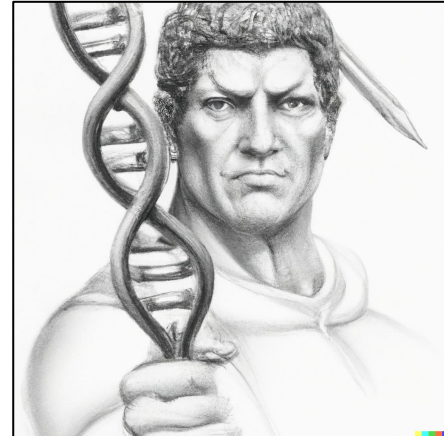
- [HMMER](#)
- [SAM](#)

Gene Prediction:

- [GENSCAN](#)
- [AUGUSTUS](#)

Protein Structure & Transmembrane Prediction:

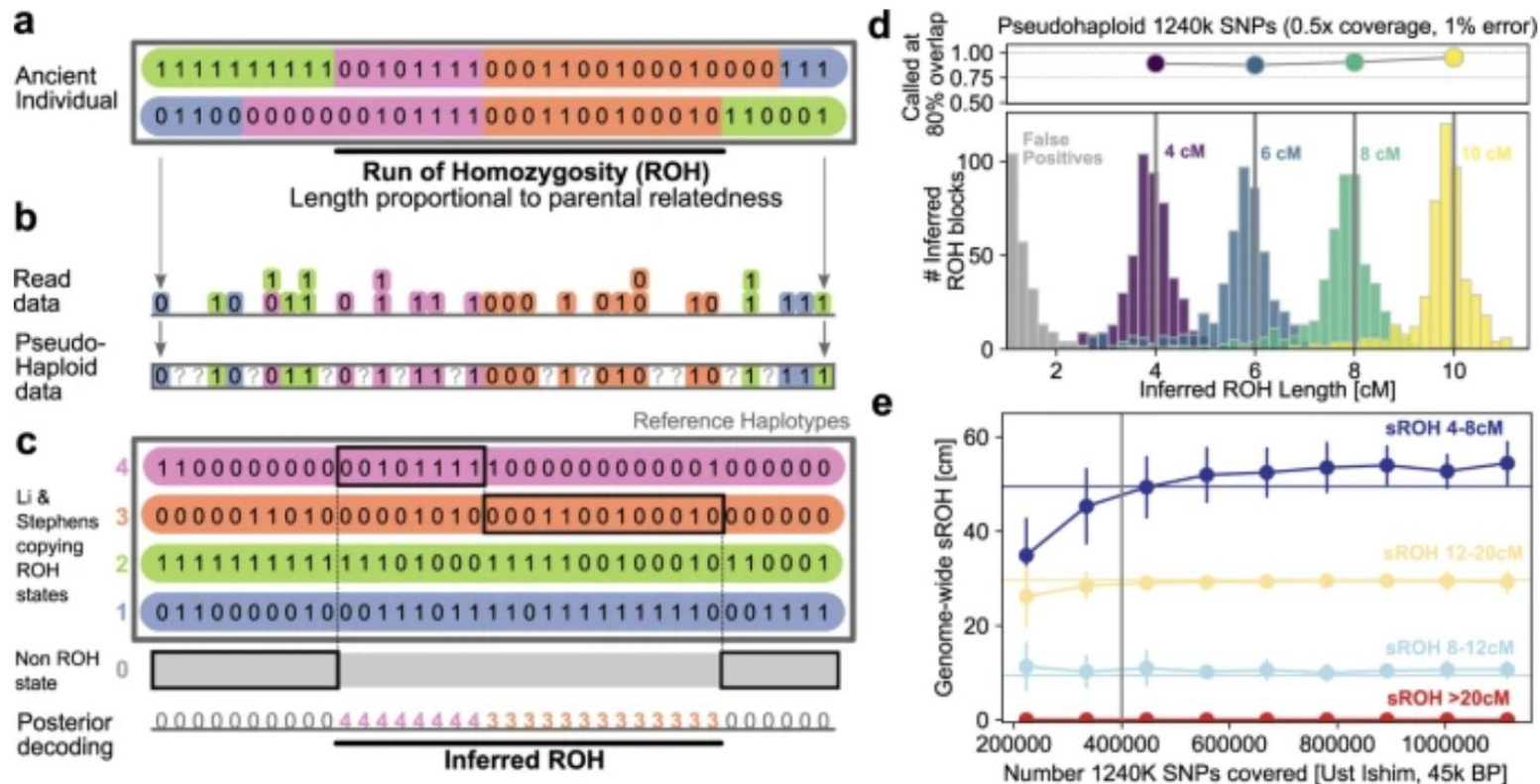
- [TMHMM](#)
- [HMMTOP](#)



HMMs in Ancient DNA Analysis

- **The problem:** Ancient DNA is degraded — most positions have noisy or missing genotype calls. We want to find runs of homozygosity: long stretches where both copies of the chromosome are identical, signaling related parents
- **Hidden state:** is this position inside an ROH or not?
- **Observation:** the genotype call at that position (homozygous or heterozygous)
- **Emission model:**
 - inside an ROH → almost all homozygous
 - outside an ROH → normal mix
- **Markov assumption:** ROHs are long, so neighboring positions almost always share the same label
- **What the HMM does:** combine noisy per-position genotype evidence with “labels come in long blocks” rule to call ROH regions along the chromosome
- **What we learn:** the number and length of ROHs reveal whether parents were related and tells us about historical population sizes

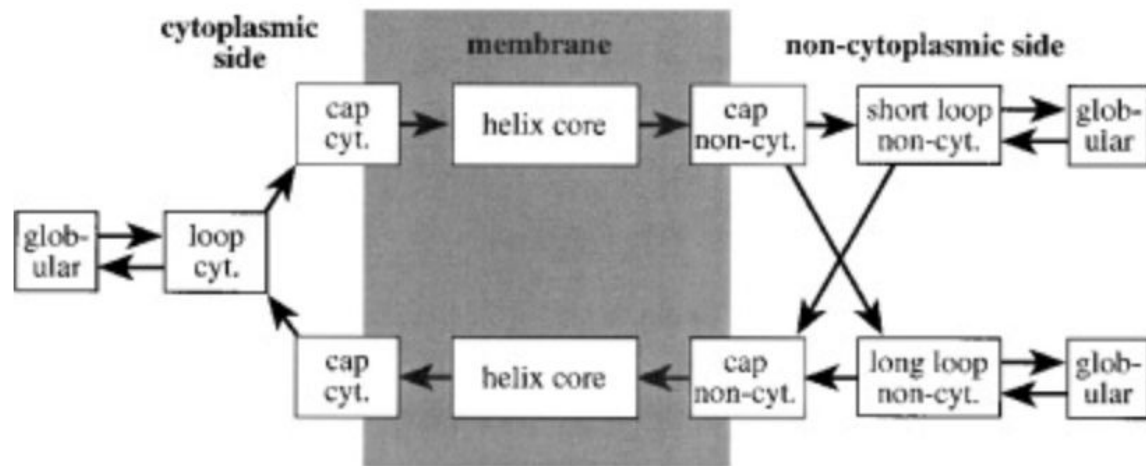
Fig. 1: Detecting runs of homozygosity using a reference panel.



HMMs in Protein Structure Prediction

- **The problem:** Some proteins are embedded in the cell membrane and snake back and forth across it. Knowing which residues sit inside the cell, in the membrane, and outside matters for signaling, drug binding, and protein function
- **Hidden state:** Is this residue inside the cell, in the membrane, or outside the cell?
- **Observation:** the amino acid at that position
- **Emission model:**
 - Inside membrane → mostly hydrophobic residues
 - inside/outside cell → mix including charged residues
- **Markov assumption:** transmembrane helices are ~20 residues long, and you can't physically jump from inside to outside without crossing membrane, so neighboring residues usually share same label
- **What the HMM does:** combine per residue amino acid evidence with “labels come in blocks and follow a topological order”
- **What we learn:** the full topology of a membrane protein — how many times it crosses the membrane, where the loops are, and which side faces the cell interior versus exterior

(a)



(b)

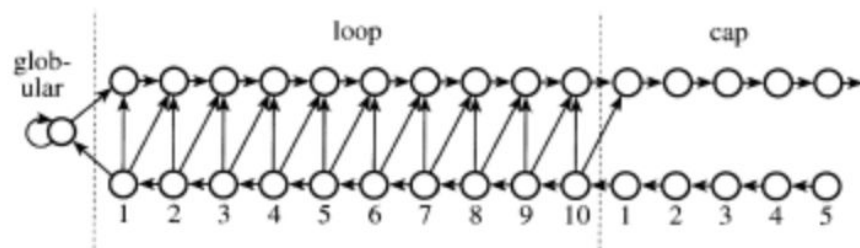


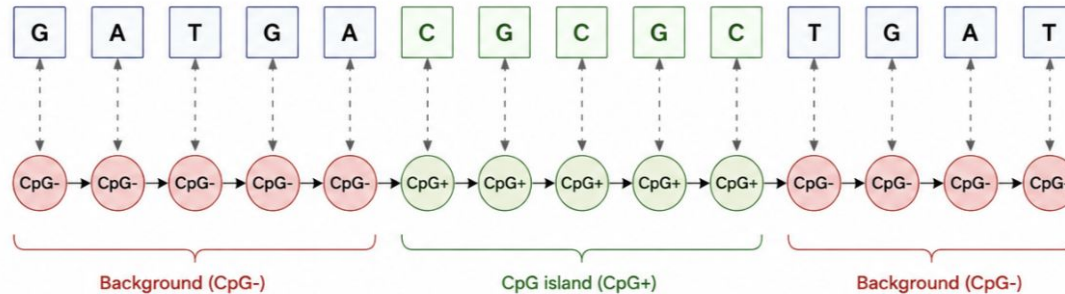
Figure 1 from Krogh et al., 2001

Demo: CpG island detection

- **CpGs:** segments of DNA with high CG dinucleotides, associated with gene regulation
- **Problem:** Identifying CpG islands helps us locate functional promoter regions of the genome, which is important in cancer, gene regulation research.

Observed DNA
sequence
(from Sanger or NGS)

Hidden state
(not observed)
(true biological state
at each position)



	CpG+	CpG-
CpG+	More Likely	Less Likely
CpG-	Less Likely	More Likely

(Deaton & Bird, 2011)

CpG island detection Model

We obtain parameters from Durbin et al. 1998, Table 3.5.

The paper defines transition matrices in and out of CpG Islands. We can transform these into our 2x16 matrix B.

$$A = \begin{bmatrix} \text{CpG+} & \text{CpG+} & \text{CpG-} \\ \text{CpG-} & 0.998 & 0.002 \\ & 0.0001 & 0.9999 \end{bmatrix} \quad \pi = \begin{bmatrix} \text{CpG+} & \text{CpG-} \\ 0.02 & 0.98 \end{bmatrix}$$

$$B^+ = \begin{bmatrix} & A & C & G & T \\ A & 0.180 & 0.274 & 0.426 & 0.120 \\ C & 0.171 & 0.368 & 0.274 & 0.188 \\ G & 0.161 & 0.339 & 0.375 & 0.125 \\ T & 0.079 & 0.355 & 0.384 & 0.182 \end{bmatrix}$$

$$B^- = \begin{bmatrix} & A & C & G & T \\ A & 0.300 & 0.205 & 0.285 & 0.210 \\ C & 0.322 & 0.298 & 0.078 & 0.302 \\ G & 0.248 & 0.246 & 0.298 & 0.208 \\ T & 0.177 & 0.239 & 0.292 & 0.292 \end{bmatrix}$$

$$B_s(XY) = P(XY \mid s) = P(X) \cdot P(Y \mid X, s) = \frac{1}{4} \cdot B_{X,Y}^s \quad \text{for } s \in \{+, -\}$$

CpG island detection Model

```
states <- c("CpG+", "CpG-")

bases <- c("A", "C", "G", "T")
symbols <- as.vector(outer(bases, bases, paste0)) # AA, AC, AG, ..., TT

startProbs <- c(0.02, 0.98)

transProbs <- matrix(c(0.998, 0.002,
                      0.0001, 0.9999), nrow = 2, byrow = TRUE)
dimnames(transProbs) <- list(states, states)
```

CpG island detection Model

```
cp_g_plus <- matrix(c(  
  0.180, 0.274, 0.426, 0.120, # A  
  0.171, 0.368, 0.274, 0.188, # C  
  0.161, 0.339, 0.375, 0.125, # G  
  0.079, 0.355, 0.384, 0.182  # T  
, nrow = 4, byrow = TRUE)
```

```
cp_g_minus <- matrix(c(  
  0.300, 0.205, 0.285, 0.210, # A  
  0.322, 0.298, 0.078, 0.302, # C  
  0.248, 0.246, 0.298, 0.208, # G  
  0.177, 0.239, 0.292, 0.292  # T  
, nrow = 4, byrow = TRUE)
```

```
rownames(cp_g_plus) <- rownames(cp_g_minus) <- bases  
colnames(cp_g_plus) <- colnames(cp_g_minus) <- bases
```

CpG island detection Model

```
emissionProbs <- matrix(0, nrow = 2, ncol = 16)
dimnames(emissionProbs) <- list(states, symbols)
```

```
for (i in seq_along(bases)) {
  for (j in seq_along(bases)) {
    sym <- paste0(bases[i], bases[j])
    emissionProbs["CpG+", sym] <- cp_g_plus[i, j] / 4
    emissionProbs["CpG-", sym] <- cp_g_minus[i, j] / 4
  }
}
```

```
emissionProbs["CpG+", ] <- emissionProbs["CpG+", ] / sum(emissionProbs["CpG+",
])
emissionProbs["CpG-", ] <- emissionProbs["CpG-", ] / sum(emissionProbs["CpG-",
])
```

CpG island detection Model

```
hmm_model <- initHMM(States = states,  
                     Symbols = symbols,  
                     startProbs = startProbs,  
                     transProbs = transProbs,  
                     emissionProbs = emissionProbs)  
  
viterbi_path <- viterbi(hmm_model, dinucs)  
  
base_states <- c(viterbi_path, viterbi_path[length(viterbi_path)])
```

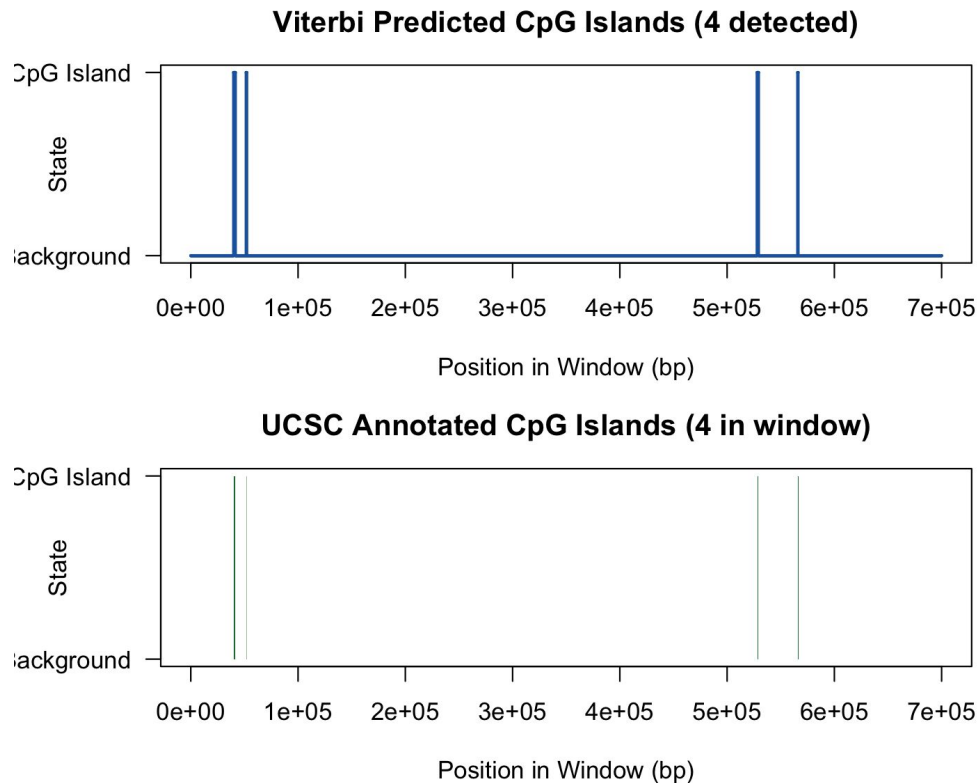
Then filter out detected runs less than 200 bps.

CpG island detection Model

We compare to the UCSC CpG Island Annotations from rtracklayer as a source of truth for some segment of Chromosome 22 in hg38.

We get a sensitivity of 94.4% and precision of 86.1%.

[Source Code](#)



Thank you!

Questions?

References

Yoon, B.-J. (2009). Hidden Markov Models and their Applications in Biological Sequence Analysis. *Current Genomics*, 10(6), 402–415.

<https://doi.org/10.2174/138920209789177575>

Krogh, A., Larsson, B., von Heijne, G., & Sonnhammer, E. L. L. (2001). Predicting transmembrane protein topology with a hidden markov model: application to complete genomes. Edited by F. Cohen. *Journal of Molecular Biology*, 305(3), 567–580.

<https://doi.org/10.1006/jmbi.2000.4315>

Ringbauer, H., Novembre, J., & Steinrücken, M. (2021). Parental relatedness through time revealed by runs of homozygosity in ancient DNA. *Nature Communications*, 12(1). <https://doi.org/10.1038/s41467-021-25289-w>

Deaton, A. M., & Bird, A. (2011). CpG islands and the regulation of transcription. *Genes & Development*, 25(10), 1010–1022.

<https://doi.org/10.1101/gad.2037511>

Saeed, U., & Usman, Z. (2019). Biological Sequence Analysis. *Exon Publications*, 55–69.

<https://doi.org/10.15586/computationalbiology.2019.ch4>